IBM University Engagement Project Submission

FITNESS BUDDY

An AI-Powered Fitness Coach

Powered by IBM Granite through IBM watsonx.ai

HEALTH & WELLNESS | EDUCATION TECHNOLOGY

Project Title & Student Details

Project Title — Fitness Buddy: AI-Powered Fitness Coach

Domain of Project — Health & Wellness / Education

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PROJECT FOCUS

Personalized home workouts, practical meal ideas, motivation and habit-building.

CORE TECHNOLOGY

IBM Cloud Lite / watsonx.ai Runtime with IBM Granite; Python and Streamlit interface.

Problem Statement No. 13 — Fitness Buddy

THE CHALLENGE

Many people struggle to maintain a healthy lifestyle because guidance can be generic, expensive, hard to fit into busy schedules, or difficult to sustain without motivation.

PROJECT OBJECTIVE

Build an accessible, conversational fitness coach that adapts practical guidance to a user's goal, fitness level, available time, equipment and food preference.

TARGET USERS

- Students and busy adults
- Beginners exercising at home
- People seeking simple everyday wellness guidance

USER PAIN POINTS

- Limited time and equipment
- Confusing or generic advice
- Inconsistent motivation
- Difficulty building repeatable habits

WHY CURRENT OPTIONS FALL SHORT

Some solutions require subscriptions or appointments; others offer rigid routines that do not adapt to a user's schedule, preferences or equipment.

REQUIRED CAPABILITIES

Personalized workouts

Home routines based on user input.

Meal ideas

Simple suggestions aligned with food preference.

Motivation

Encouragement and practical restart tips.

Habit-building

Daily checklists and a weekly consistency view.

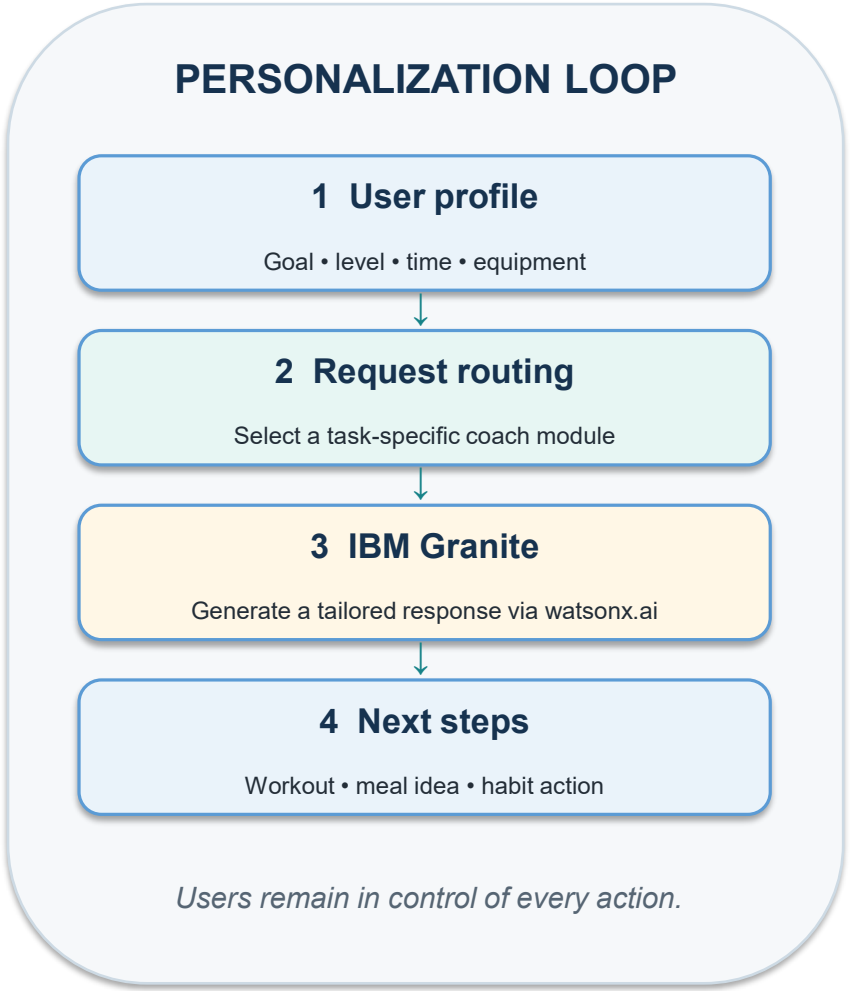
Proposed Solution — Fitness Buddy

SOLUTION OVERVIEW

Fitness Buddy is a browser-based conversational assistant built with Python and Streamlit. It uses IBM Granite through watsonx.ai to provide practical, profile-aware responses for common fitness and wellness tasks.

HOW IT DELIVERS VALUE

- Uses the user's stated level, goal, time, equipment and food preference.
- Provides actionable responses in one simple interface.
- Stores habit checklist entries locally and summarizes recent consistency.
- Routes queries to the right fitness specialist.



Technology Used

IBM Cloud Lite / Runtime

Provides the IBM cloud environment and managed watsonx.ai Runtime for model inference, subject to plan and quota.

IBM Granite 4 H Small

Primary conversational foundation model: ibm/granite-4-h-small.

IBM watsonx.ai

Hosts the inference endpoint used by the application to request Granite responses.

Python

Implements application logic, profile context, intent routing, validation and helper modules.

Streamlit + Flask

Streamlit is the primary browser UI; Flask exposes an alternative REST API entry point.

Local JSON storage

Persists habit checklist data locally in data/habits.json for the prototype.

Development note: IBM Bob was used as an AI-assisted coding environment. It is a development tool, not a runtime dependency of the submitted application.

Application Components Used

Profile Input

Captures user-selected fitness level, goal, available time, equipment and dietary preference.

Intent Router

Identifies whether the request is about a workout, meal, motivation/habit or general fitness.

Specialist Modules

Task-specific prompt and helper logic for workouts, nutrition, motivation and wellness.

Granite Connector

Reads watsonx.ai credentials securely and calls the selected IBM Granite model.

Habit Tracker

Records daily checklist selections in local JSON storage.

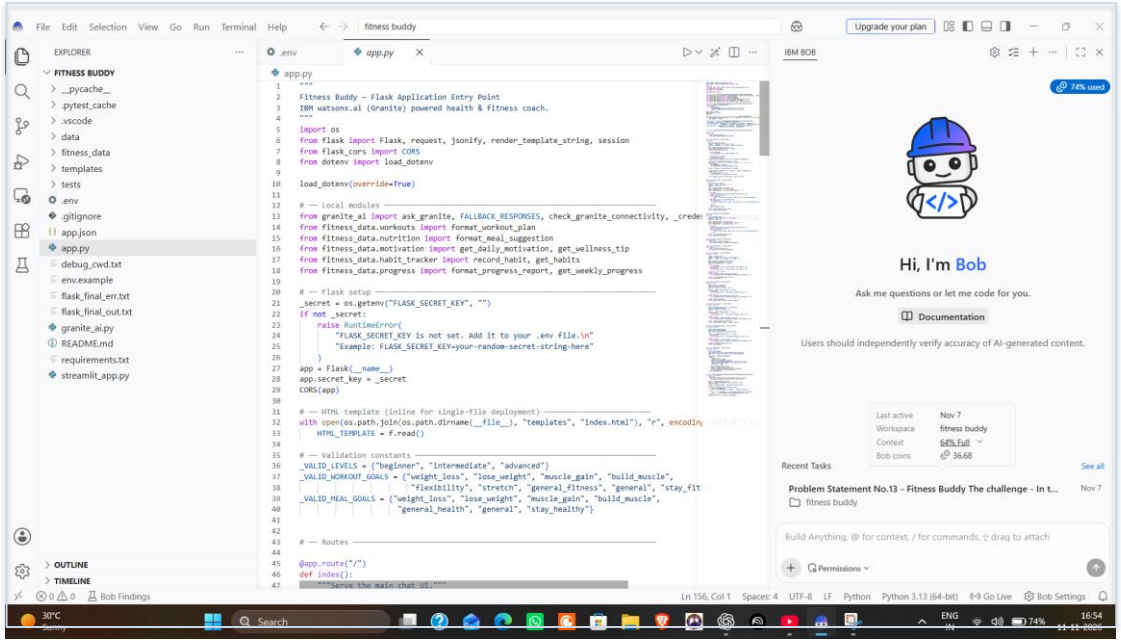
Progress Dashboard

Summarizes habit completion over recent days to support consistency.

Key project files: streamlit_app.py • granite_ai.py • fitness_data/workouts.py • nutrition.py • motivation.py • habit_tracker.py • progress.py

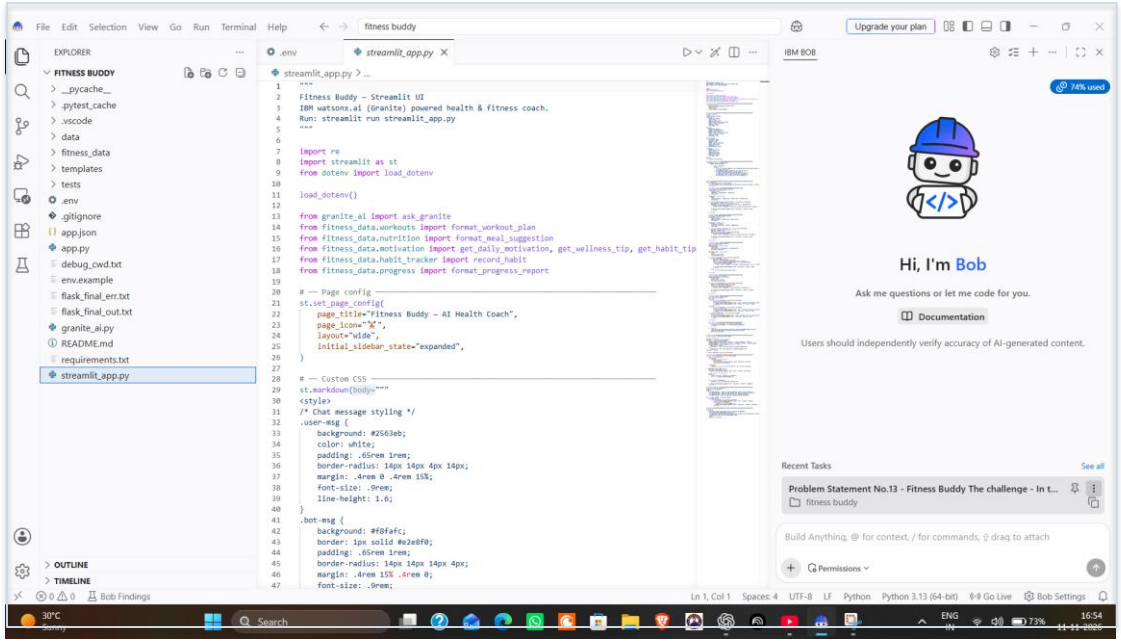
Development Process Using IBM Bob

Actual project files open in the IBM Bob IDE during implementation.



Backend implementation — app.py

Flask routes, request handling, validation and the integration points used by the Fitness Buddy application.

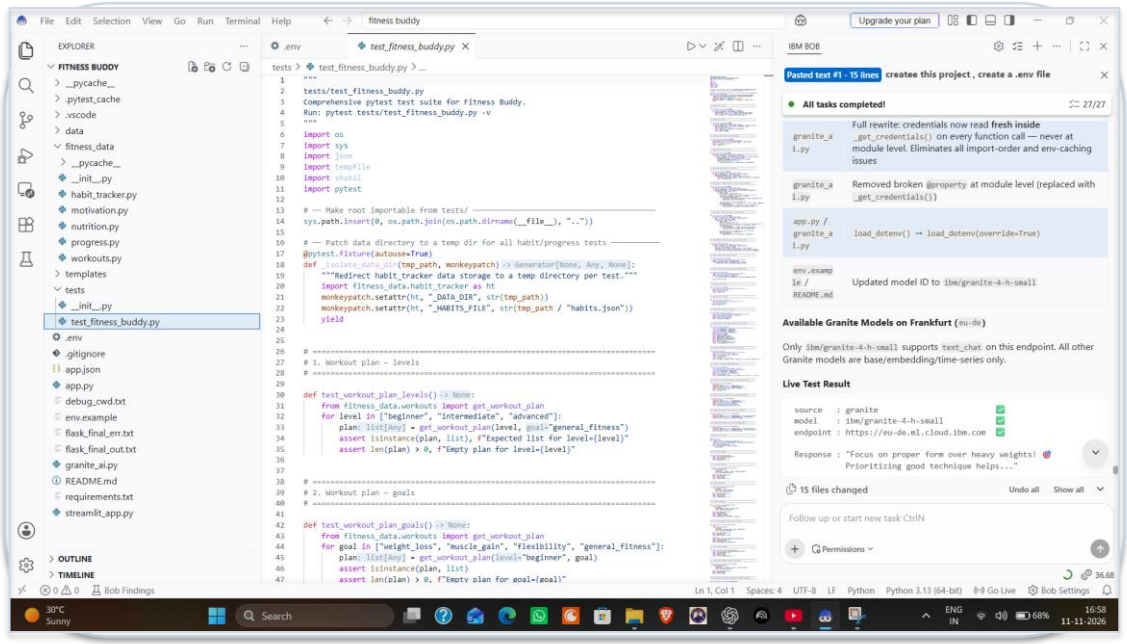


Frontend implementation — streamlit_app.py

Streamlit page setup, interface layout, chat styling and connections to the workout, nutrition, habit and progress modules.

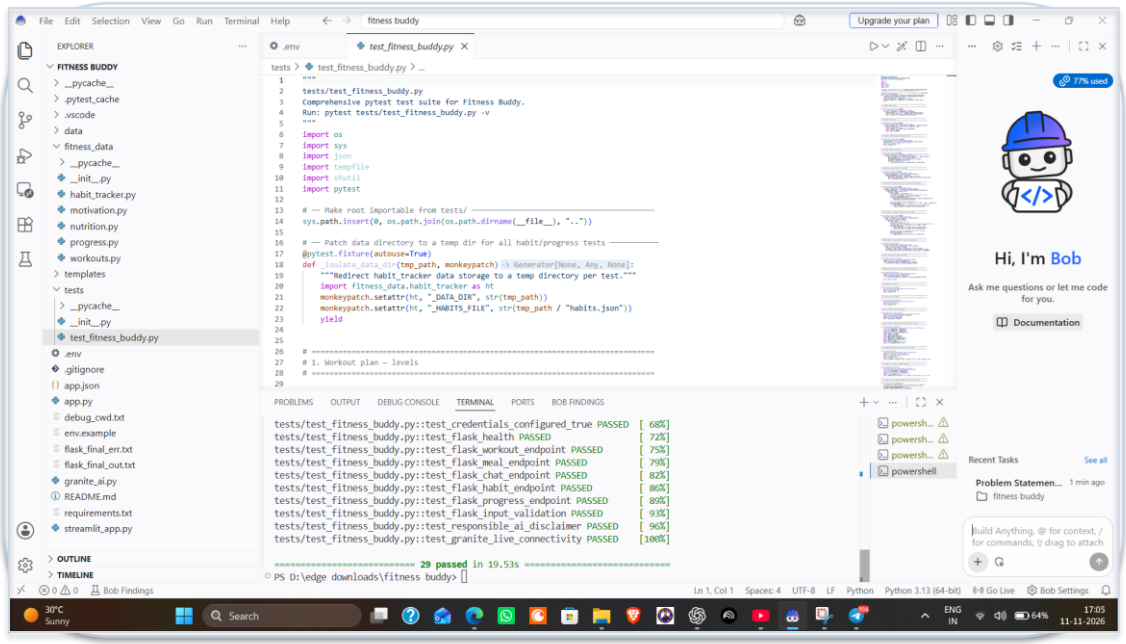
IBM Bob Implementation & Verification

Development-task completion, live IBM Granite inference and automated test evidence.



IBM Bob task completion & Granite live test

Bob reports 27/27 development tasks completed. The live test panel shows source: granite, model ibm/granite-4-h-small and the Frankfurt endpoint.

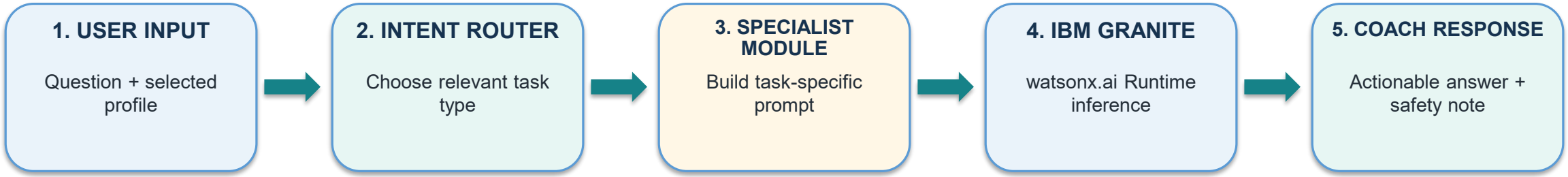


Automated testing with pytest

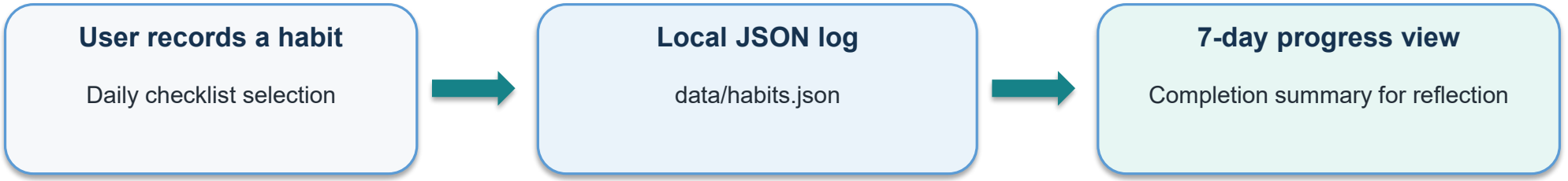
The local test run collected 29 tests and finished with 29 passed in 19.53 seconds.

Agent Workflow

REQUEST-TO-RESPONSE FLOW

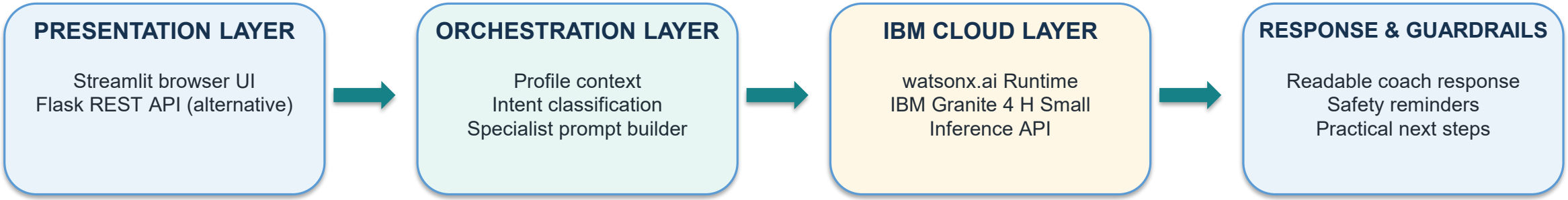


HABIT & PROGRESS LOOP



The prototype uses bounded task routing and user-triggered actions. It does not run autonomous background agents or retrain the model from user history.

Architecture Blueprint



LOCAL DATA LAYER

data/habits.json
Daily habit records and weekly summary.
Stored locally in the MVP.

SECURE CONFIGURATION

WATSONX_API_KEY • PROJECT_ID • URL • MODEL_ID
Loaded from .env / secrets.
Never committed to GitHub.

PROJECT BOUNDARY

Text-based prototype • sample meal guidance
No wearable, voice or medical-record integration yet.

Primary runtime path: Streamlit → task routing → IBM watsonx.ai Runtime → IBM Granite → formatted response. Flask provides an alternative API entry point.

Role of Agentic AI in Fitness Buddy

INTENT-AWARE ACTION

Recognizes the request type and routes it to a workout, nutrition, motivation/habit or general fitness prompt.

PERSONALIZATION

Uses the user-stated goal, fitness level, available minutes, equipment and food preference as prompt context.

SPECIALIST TASK HANDLING

Each task module shapes the instruction for IBM Granite instead of using one generic prompt for every request.

CONTINUITY THROUGH HABITS

Local habit records and a 7-day summary help users reflect on consistency and choose a next step.

CONTROLLED, USER-LED SUPPORT

The user decides what to follow; the prototype does not autonomously prescribe or execute health actions.



Understand → Plan → Respond → Track

Responsible AI & Prototype Boundaries

NO MEDICAL DIAGNOSIS

Fitness Buddy is an educational assistant. It does not diagnose conditions, treat illness or replace qualified professionals.

SAFE FITNESS GUIDANCE

Avoids extreme routines and encourages users to stop if they experience serious symptoms such as chest pain, faintness or severe breathlessness.

NUTRITIONAL CAUTION

Meal suggestions are general ideas. Calorie and macro values are estimates, not medical diet prescriptions.

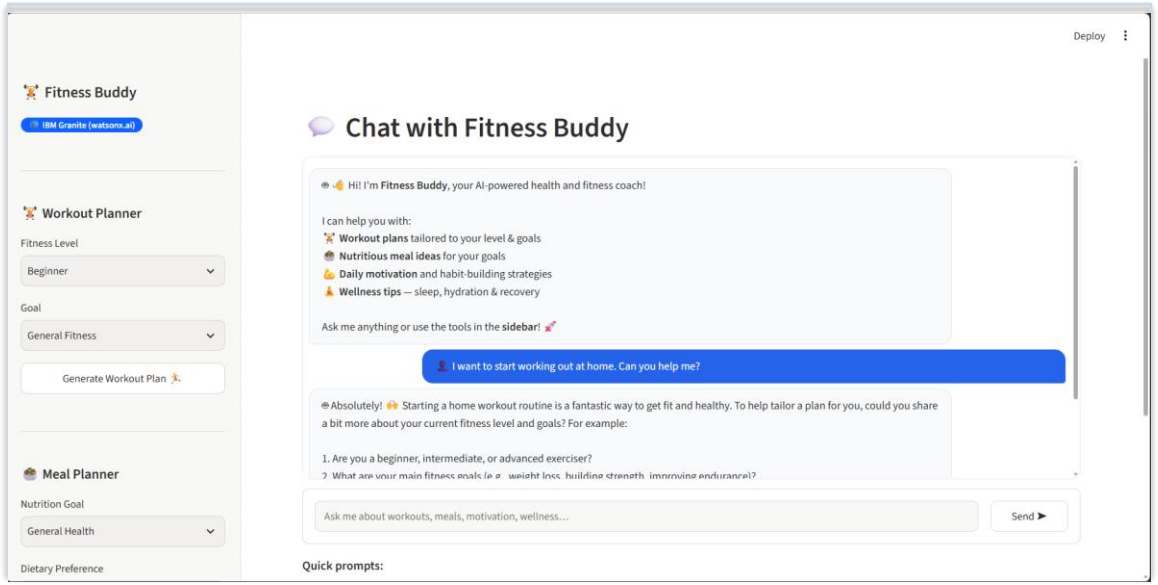
PRIVACY & USER CONTROL

Profile details are user-provided. Habit data is stored locally for this prototype; credentials are kept outside the public repository.

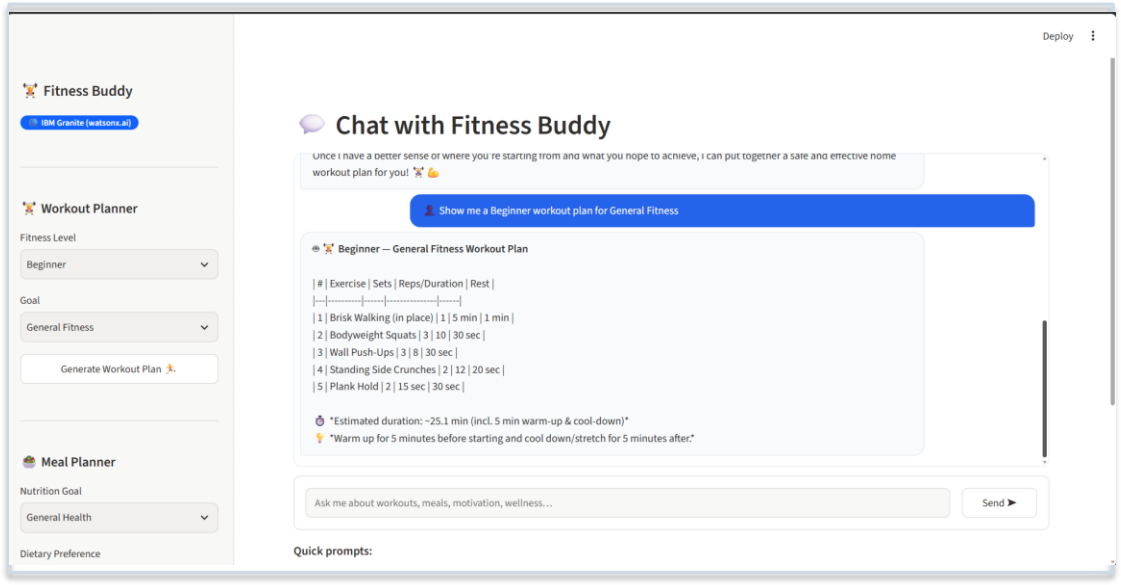
If serious symptoms occur: stop exercising and seek appropriate medical help. Users should consult a qualified professional for medical conditions, injuries or special dietary needs.

Demo / Simulation — Actual Application Screens

The screenshots below are from the running local Streamlit prototype.



1. Chat interface and feature navigation



2. Beginner workout plan generated in the UI

DEMO INPUT

Beginner • General Fitness • No equipment

USER INTERACTION

Ask a natural-language question or use the sidebar planner.

VISIBLE OUTPUT

The app displays a structured routine and coaching guidance.

Prototype Testing & Results

29 / 29

Automated tests passed

LIVE IBM GRANITE INFERENCE VERIFIED

Model: ibm/granite-4-h-small | Endpoint: Frankfurt (eu-de) | API source: granite

TEST COVERAGE REPORTED BY THE PROJECT AUDIT

Workout plans

Levels, goals, duration bounds

Meal planner

Diet filters and calorie summary

Habits & progress

Record, retrieve and 7-day summary

API & validation

Endpoints, invalid input handling

Safety

Medical disclaimer and credential guards

Granite connector

Fallback separation and live inference check

Test figures reflect the IBM Bob audit report and local project run. Re-run the test suite after any further code change.

Novelty and Uniqueness

Fitness Buddy combines conversational coaching with small, practical tools that help users move from a question to a realistic next step.

01 TASK-SPECIFIC ROUTING — Requests are directed to specialist workout, nutrition, motivation/habit or general fitness instructions.

02 CONSTRAINT-AWARE ADVICE — Recommendations reflect the user's selected goal, skill level, time, equipment and food preference.

03 EVERYDAY USABILITY — A simple Streamlit interface brings chat and common planning actions into one browser view.

04 CONSISTENCY SUPPORT — A local habit checklist and 7-day summary emphasize repeatable habits instead of perfection.

05 RESPONSIBLE SCOPE — The prototype is explicit about its educational role and avoids diagnosis or extreme recommendations.

Differentiator: a lightweight, student-built prototype focused on approachable personalization and habit continuity.

GitHub Repository & Submission

PUBLIC GITHUB REPOSITORY

https://github.com/keshavpareek603-star/fitness-buddy-agent

REPOSITORY CONTENT

- Python application source
- fitness_data, templates and tests
- app.json, README.md and requirements.txt
- env.example and .gitignore
- Problem_statement_Fitness_Buddy.pdf

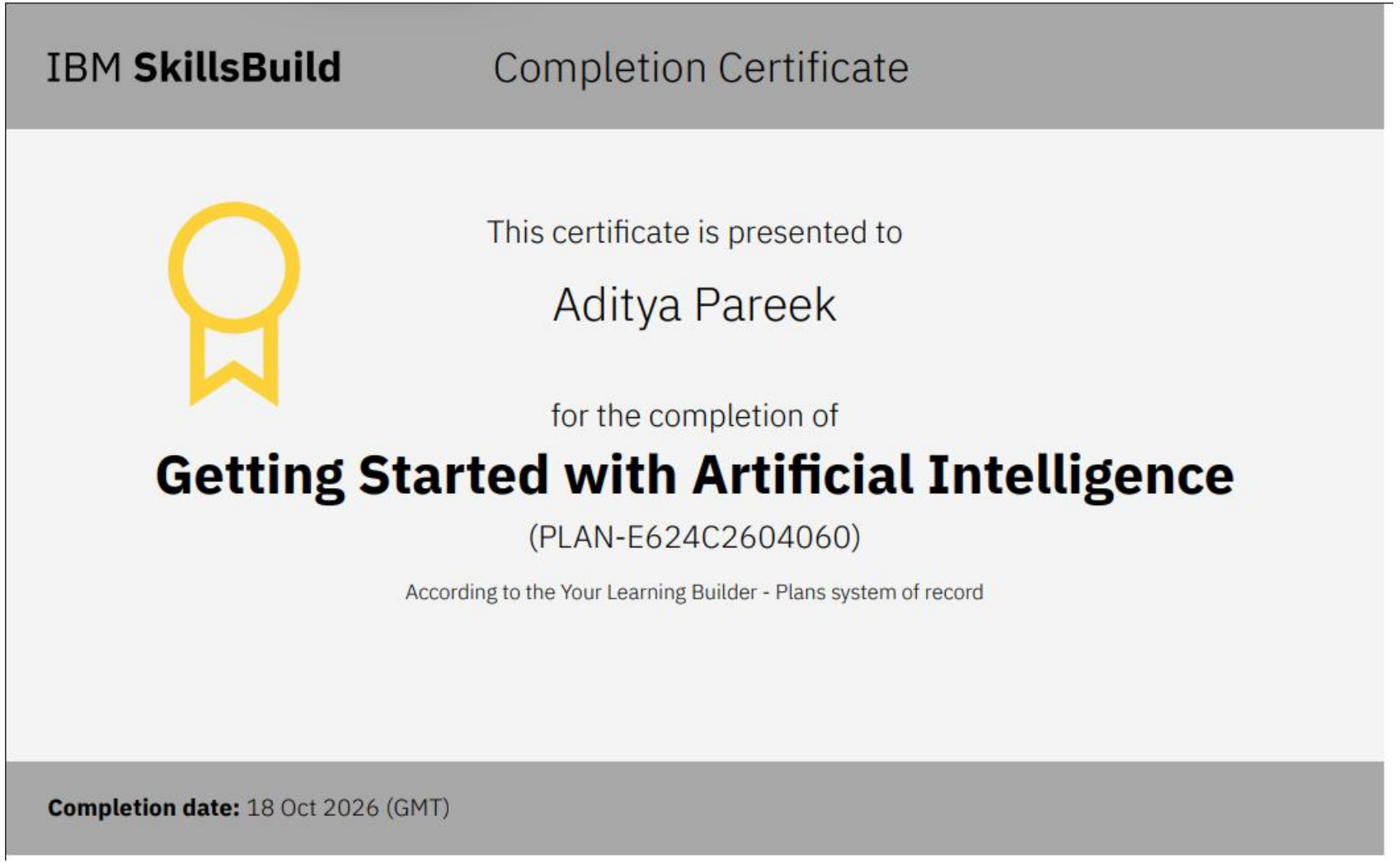
PUBLISHING & SECURITY

- Repository visibility: Public
- .env and API secrets are excluded
- Personal habit data is excluded
- Final PPTX must be added to this repository after completion
- Use the same final PPTX in the official submission portal

Future Scope

Voice and multilingual access	Add speech input/output and support regional languages to improve accessibility.
Wearable integration	With user consent, use selected activity or sleep summaries from compatible devices.
Reviewed wellness knowledge	Add a curated knowledge base with reliable sources, dates and citations for general wellness information.
Richer habit insights	Offer opt-in reminders, streaks, weekly reflections and user-controlled export.
Secure cloud deployment	Consider authenticated cloud storage only after consent, retention and data-deletion controls are designed.

Certificates Earned — IBM SkillsBuild (Getting Started with Artificial Intelligence)



IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Aditya Pareek

for the completion of

Getting Started with Artificial Intelligence

(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of record

Completion date: 18 Oct 2026 (GMT)

Certificate Earned — IBM Bob (Lab: Troubleshoot Your Code Using IBM Bob)



Thank You!

Fitness Buddy — AI-Powered Fitness Coach

Questions & feedback are welcome.